

AWS Glue Interview Questions and Answers

A short set of commonly asked interview questions covering AWS Glue fundamentals, the Data Catalog, crawlers, ETL job concepts, and how Glue compares to other AWS data-processing services.

Q1. What is AWS Glue?

AWS Glue is a serverless data integration service used to discover, prepare, and combine data for analytics, machine learning, and application development. It provides ETL (extract, transform, load) capabilities without requiring you to provision or manage servers.

Q2. What is the Glue Data Catalog?

The Glue Data Catalog is a persistent metadata store. It holds table definitions, schema information, and location details for data sources such as S3, RDS, and Redshift. Other AWS services like Athena, Redshift Spectrum, and EMR can use it as a central metadata repository.

Q3. What is a Glue Crawler?

A crawler connects to a data source, infers the schema of the data, and automatically creates or updates table definitions in the Data Catalog. Crawlers can be scheduled to run periodically to keep metadata up to date as new data arrives.

Q4. What is a DynamicFrame and how does it differ from a Spark DataFrame?

A DynamicFrame is Glue's own data abstraction, similar to a Spark DataFrame but designed to handle semi-structured data with inconsistent or evolving schemas. It tracks schema inconsistencies per record and offers built-in transforms like ResolveChoice and Relationalize; it can be converted to and from a Spark DataFrame when needed.

Q5. What are Job Bookmarks in Glue?

Job bookmarks let a Glue ETL job keep track of which data has already been processed. On subsequent runs, the job only processes new or changed data, avoiding duplicate processing of records that were already handled in earlier runs.

Q6. What is the difference between Glue and Amazon EMR?

Glue is a fully serverless, managed ETL service best suited for simpler, event-driven, or scheduled transformations with minimal operational overhead. EMR gives you a managed Hadoop/Spark cluster with full control over configuration, instance types, and installed frameworks, which suits larger or more customized big-data workloads.

Q7. What are Glue triggers and workflows?

Triggers start Glue jobs or crawlers based on a schedule, an event, or the completion of another job. Workflows let you group multiple jobs, crawlers, and triggers into a single orchestrated pipeline, so you can visualize and manage dependencies between ETL steps.

Q8. How does Glue handle schema evolution?

Crawlers can detect schema changes between runs and update the Data Catalog accordingly. Within ETL code, transforms such as ResolveChoice let you explicitly decide how to reconcile columns that have inconsistent types across records.

Q9. What is Glue Studio?

Glue Studio is a visual interface for creating, running, and monitoring ETL jobs. It lets you build a job as a visual graph of sources, transforms, and targets, and it auto-generates the underlying Spark or Python code.